StressEar-AI: An Explainable, Public-Data AI Framework for Occupational Stress, Tinnitus, and Hearing Outcomes

A Study Protocol and External-Validation Plan

Geunsik Lim[1] and Hyun Jo[2]

[1]Sungkyunkwan University, Republic of Korea; leemgs@g.skku.edu

[2]Ajou University School of Medicine, Republic of Korea; joehyun@ajou.ac.kr

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Abstract

Purpose: We present StressEar-AI (Stress–Ear Artificial Intelligence), a research framework and pre-analysis study protocol for characterizing the associations among occupational (psychosocial) stress, tinnitus, and hearing outcomes in employed middle-aged adults, using openly accessible population datasets and clinician-verified artificial intelligence (AI). The framework is deliberately designed to avoid causal overclaiming: occupational stress is treated as a psychosocial exposure that may be associated with tinnitus burden, help-seeking behavior, and hearing-related outcomes, rather than as a proven direct cause of cochlear injury.

Method: We select the Korea National Health and Nutrition Examination Survey (KNHANES) as the primary development dataset and the U.S. National Health and Nutrition Examination Survey (NHANES) as the geographic external-validation dataset. Both resources support survey-weighted, population-level analyses of pure-tone audiometry, self-reported tinnitus or hearing difficulty, noise exposure, and health covariates. We prespecify weighted regression models and two calibrated, interpretable prediction targets (tinnitus status and audiometric hearing loss), evaluated with discrimination, calibration, and subgroup-fairness metrics, and reported per TRIPOD+AI and STROBE. An optional prospective clinical extension at Ajou University Hospital adds deidentified audiograms and otolaryngology records for external clinical validation and for studying treatment timing in sudden sensorineural hearing loss. The AI component combines transparent statistical models, gradient-boosted trees, and an open medical language model (e.g., MedGemma) for schema-constrained extraction from deidentified clinical narratives; AI outputs are restricted to feature extraction, calibrated risk estimates, and guideline-linked explanations, and are governed by a deterministic red-flag referral layer.

Expected Contribution: StressEar-AI contributes a reusable public-data baseline, a cross-national external-validation workflow, a clinician-review and safety protocol, model cards, and fully open code, so that researchers worldwide can reproduce, criticize, and extend the analyses with their own cohorts. The framework prioritizes reproducibility, fairness, clinical safety, and broadly shared benefit.

Conclusions: Public datasets can support robust association analyses and can benchmark explainable AI methods, but prospective clinical cohorts remain necessary to evaluate treatment timing, recovery, and individual prognosis. StressEar-AI offers a transparent path from open epidemiological evidence to clinically grounded, explainable audiology AI.

Keywords: tinnitus; hearing loss; audiometry; occupational stress; sudden sensorineural hearing loss; KNHANES; NHANES; explainable AI; open medical language models; MedGemma; TRIPOD+AI; reproducibility; audiology.

1 Introduction

Tinnitus and hearing loss are among the most common auditory health concerns worldwide, affecting communication, sleep, emotional well-being, work participation, and quality of life. The World Health Organization estimates that more than 1.5 billion people live with some degree of hearing loss, a number projected to rise substantially by 2050 (World Health Organization, 2021). Tinnitus alone affects an estimated 10–15% of adults, and a clinically meaningful subset experiences bothersome tinnitus that interferes with concentration and mood (Tunkel et al., 2014; McCormack et al., 2016). In employed middle-aged adults, tinnitus frequently emerges in a context of psychosocial stress, long working hours, sleep disruption, occupational or recreational noise, cardiometabolic risk, and delayed help-seeking. Because these factors are temporally and biologically intertwined, simple causal narratives are unreliable and can mislead both patients and clinicians.

1.1 Motivating clinical observation

The framework is motivated by a common and instructive clinical trajectory, presented here as an anonymized composite rather than as identifiable patient data. A worker in the late 40s to early 50s first experiences unilateral tinnitus in one ear during a period of heavy workload. Reasoning intuitively, the person “rests” the affected ear by shifting phone calls and earphone use to the contralateral side; over roughly nine months the tinnitus resolves.

About a year later, the previously unaffected ear develops sudden aural fullness and a "pop" during a stressful workday, followed within days to weeks by reduced audibility of soft sounds. Lacking awareness of the narrow treatment window for sudden sensorineural hearing loss (SSNHL), the person delays evaluation for roughly two months; by the time intratympanic and systemic steroid therapy is attempted, recovery is incomplete. This trajectory highlights two clinically actionable points that StressEar-AI (Stress–Ear Artificial Intelligence) is built around: (i) stress-associated tinnitus is common and often self-limiting, but (ii) sudden or rapidly progressive hearing loss is a medical urgency in which delayed care—not stress per se—may drive irreversible impairment. Attributing acute hearing symptoms to "stress" without audiological and otolaryngological assessment is therefore hazardous.

1.2 Conservative premise

StressEar-AI begins from a conservative and testable premise: occupational stress may be *associated* with tinnitus and hearing outcomes, but public cross-sectional datasets are primarily suited to estimating associations, not to proving that stress directly causes tinnitus or sensorineural hearing loss. This distinction is central to the framework's clinical safety posture and to its statistical claims.

1.3 Aims and hypotheses

The framework has four aims, each with prespecified, falsifiable hypotheses:

1. **Association (development).** Estimate weighted associations between perceived stress and (a) tinnitus status and (b) audiometric hearing loss in KNHANES, adjusting for noise exposure and health covariates. *H1: A positive association between perceived stress and tinnitus persists after adjustment; the association with audiometric thresholds is weaker and may attenuate to non-significance after adjusting for noise and age.*
2. **Transportability (external validation).** Test whether harmonized feature definitions

and prediction models developed in KNHANES retain discrimination and calibration in NHANES. *H2: Discrimination degrades modestly but calibration can be restored by recalibration, supporting cross-national transportability of the tinnitus model more than the hearing-loss model.*

3. **Explainable, safe AI.** Evaluate whether an open medical language model can extract audiology-relevant fields from deidentified narratives at clinician-verified accuracy while a deterministic red-flag layer reliably routes SSNHL and neurologic warning signs to urgent referral. *H3: Schema-constrained extraction reaches high human-verified agreement with a low unsupported-output rate, and the safety layer achieves near-complete red-flag recall.*

4. **Open, shared benefit.** Provide an openly licensed, reproducible pipeline (code, data dictionary, model cards, prompts) that lowers the barrier for researchers to reproduce analyses and add local cohorts.

1.4 Scope and positioning

This manuscript updates the prior StressEar-AI concept for the *American Journal of Audiology*. The revised emphasis is clinical audiology: pure-tone thresholds, tinnitus burden, speech-in-noise implications, unilateral and asymmetric hearing difficulty, hearing rehabilitation, and safe, timely referral for urgent symptoms. AI is treated throughout as an enabling research tool, not as a replacement for clinical judgment. As a pre-analysis protocol, the manuscript reports the design, safety governance, and reproducible scaffold; it does not report definitive epidemiological findings, which await the analyses specified herein.

2 Related Work

2.1 Epidemiology of tinnitus, hearing loss, and stress

Population studies have repeatedly shown that tinnitus and hearing loss often coexist, but their relationships are heterogeneous. Analyses using KNHANES have estimated tinnitus prevalence and associated factors in Korea and demonstrated that nationally representative data can support clinically meaningful tinnitus research (Park et al., 2014; Joo et al., 2015). NHANES has similarly enabled U.S. studies of audiometry, occupational noise exposure, tinnitus, and mental-health correlates (Hoffman et al., 2020; Chakrabarty et al., 2024; Mahboubi et al., 2013). Reviews of tinnitus epidemiology emphasize wide prevalence estimates driven by heterogeneous case definitions, underscoring the need for harmonized variables when pooling or comparing across surveys (McCormack et al., 2016). Evidence linking psychological stress to tinnitus onset and severity is consistent but largely observational; longitudinal occupational cohorts suggest stress is associated with incident tinnitus, yet residual confounding by noise, sleep, and cardiometabolic factors is difficult to exclude (Baigi et al., 2011).

2.2 Beyond the pure-tone average

Recent audiology work highlights the need to measure more than the pure-tone average. Executive-control studies in individuals with tinnitus suggest that cognitive burden and attentional control may be relevant to perceived functional impact (American Journal of Audiology, 2026a). Speech-perception-in-noise and hearing-aid signal-management studies demonstrate that real-world listening difficulties can persist even when threshold-based metrics appear near-normal (American Journal of Audiology, 2026b,c). These findings motivate StressEar-AI to include tinnitus severity, communication difficulty, occupational participation, and listening effort as outcomes alongside audiometric thresholds.

2.3 Clinical pathways and the treatment window

Clinical practice guidelines frame the endpoints StressEar-AI must respect. The clinical practice guideline for tinnitus distinguishes bothersome from non-bothersome tinnitus and prioritizes education, cognitive-behavioral strategies, and sound therapy over pharmacologic cures (Tunkel et al., 2014). Critically, the guideline for sudden hearing loss defines SSNHL as a medical urgency: prompt audiometric confirmation and early corticosteroid therapy within roughly two weeks offer the best chance of recovery, and delayed presentation is associated with worse outcomes (Chandrasekhar et al., 2019). This "golden-window" principle is the clinical rationale for the deterministic red-flag layer described in Section 5.

Prior work involving Hun Yi Park and colleagues is especially relevant for the clinical expansion of this study. Sound-localization findings in unilateral hearing-aid users show that the side of hearing impairment may influence spatial-hearing interpretation (Ha et al., 2022). Early-management research in traumatic benign paroxysmal positional vertigo, although not a tinnitus study, underscores the importance of rapid identification and pathway-based care for acute otologic and vestibular symptoms (Kim et al., 2022). Cochlear-implant research using a CT-derived basal turn–facial ridge angle illustrates how explainable anatomical variables can support prediction of residual hearing preservation (Kim et al., 2021). Together, these studies support a StressEar-AI design that separates screening, referral, treatment, and rehabilitation endpoints.

2.4 Open medical AI and reporting standards

Open medical AI models are increasingly available for research use. MedGemma is a collection of open models optimized for medical text and image comprehension, positioned as a starting point for health-AI development rather than a validated diagnostic device (Google Research, 2025; Google Health AI, 2026). For StressEar-AI, this implies constraining the model to schema extraction and evidence-linked explanations, with clinician verification and external validation before any clinical use. Methodologically, the framework adheres to established reporting standards for observational and prediction research—STROBE for observational analyses (von Elm et al., 2007) and the TRIPOD+AI statement for AI-based prediction models (Collins et al., 2024)—so that reported models are transparent, comparable, and auditable.

3 Public Dataset Selection

Dataset selection was governed by four requirements for the primary association analyses: (i) survey-weighted population representativeness; (ii) objective pure-tone audiometry; (iii) tinnitus and/or hearing-difficulty items; and (iv) covariates for noise exposure, psychosocial stress, and general health. Only KNHANES and NHANES jointly satisfy all four at population scale; other public resources are retained as optional secondary datasets. Table 1 summarizes the selection, and Table 2 maps target variables to each source.

3.1 Primary Development Dataset: KNHANES

KNHANES is selected as the primary development dataset because it is nationally representative for Korea and has previously supported tinnitus and hearing-loss analyses (Park et al., 2014; Joo et al., 2015). Its advantages include Korean population relevance, complex-survey weights, pure-tone audiometry in applicable cycles (otologic examination cycles from 2009–2012 onward), tinnitus and hearing-difficulty questionnaire items in selected cycles, health and behavioral covariates, and psychosocial or stress-related self-report variables (e.g., perceived-stress and depressed-mood items). It also provides a realistic bridge to future Korean hospital cohorts, including the planned Ajou University Hospital extension. Because item wording and audiometry availability differ across cycles, the analysis prespecifies a per-cycle codebook review and restricts pooling to cycles with comparable definitions.

The limitations are equally important. KNHANES is not a workplace cohort, and the audiometric analyses are cross-sectional. It can therefore evaluate *associations* among perceived stress, work-related variables, tinnitus, and audiometric outcomes, but cannot by itself establish that occupational stress *caused* tinnitus or hearing loss. Treatment timing and recovery after sudden hearing loss are not captured.

3.2 External-Validation Dataset: NHANES

NHANES is selected as the primary external-validation dataset because it provides openly downloadable U.S. public-use files, pure-tone audiometry and tympanometry in selected cycles (e.g., 2011–2012, 2015–2016, 2017–March 2020), occupational and recreational noise-exposure items, tinnitus and hearing questionnaires in selected cycles, and extensive demographic and health covariates. NHANES enables testing whether feature definitions and risk models derived from KNHANES retain discrimination and calibration across population, language, health-system, and occupational contexts. As with KNHANES, complex-survey weights, strata, and primary sampling units must be honored, and cycle-specific age eligibility for the audiometry module must be respected.

3.3 Optional Secondary Datasets

Open audiology repositories may be used for targeted secondary experiments: the Oldenburg Hearing Health Repository and Zenodo auditory-test datasets for audiometric model benchmarking and audiogram-based clustering; OpenNeuro tinnitus neuroimaging and fNIRS datasets for an optional neurophysiological substudy of tinnitus subtypes; and Mendeley/Zenodo tinnitus EEG datasets for explainable-biomarker exploration. These resources are not primary because they typically lack the joint combination of population weights, tinnitus items, audiometry, occupational factors, and general-health covariates required for the main survey-weighted analyses. Any such dataset is used only after verifying its license, consent basis, and permission for secondary research use.

3.4 Prospective Clinical Extension (Ajou University Hospital)

A prospective otolaryngology/audiology cohort is specified to address what public cross-sectional data cannot: treatment timing, SSNHL recovery trajectories, tinnitus burden over time, and hearing-aid rehabilitation. This cohort contributes deidentified audiograms and clinical narratives for clinical external validation and provides the substrate for the AI extraction and safety evaluations. Raw clinical data are not redistributed; only derived, deidentified, or synthetic artifacts are shared through the open repository, subject to institutional review board (IRB) approval and data-governance review.

4 Methods

This pre-analysis protocol is reported following STROBE for the observational analyses (von Elm et al., 2007) and TRIPOD+AI for the prediction models (Collins et al., 2024); the prospective clinical extension will follow SPIRIT. A completed reporting checklist accompanies the repository.

4.1 Study Design

The work has three stages. **Stage 1** analyzes KNHANES as the primary public-data development cohort (association estimation and model development). **Stage 2** evaluates cross-national transportability in NHANES (geographic external validation). **Stage 3** prepares a prospective clinical extension at Ajou University Hospital for clinical external validation, treatment-timing analysis in SSNHL, and evaluation of the AI extraction and safety components. Stages 1–2 use only deidentified public-use files; Stage 3 proceeds only under IRB approval.

4.2 Participants and Eligibility

For KNHANES and NHANES, eligible records include adults within the age range covered by the audiometry modules, with emphasis on middle-aged employed participants (approximately 40–65 years) when variables permit. Records are included when they have valid survey weights and non-missing values for the outcome under analysis; a complete-case sensitivity analysis is compared against multiply imputed analyses (below). Conductive patterns are flagged and, where tympanometry is available, excluded in a sensitivity analysis so that sensorineural-predominant thresholds can be examined separately.

4.3 Variable Harmonization

A published harmonization layer maps age, sex, education, occupation, employment status, perceived stress, sleep duration, smoking, alcohol, cardiometabolic disease, noise exposure (occupational and recreational, including firearm use), tinnitus status and bother, hearing difficulty, hearing-aid use, and pure-tone thresholds into a common schema (Table 2). The mapping is stored as a human-readable, version-controlled file; every derived variable records its source items, cycle, and transformation. Analysis code preserves survey cycle, sampling strata, cluster identifiers, and weights so that all estimates are design-consistent.

4.4 Exposure and Outcomes

The primary exposure is stress-related self-report, operationalized from each dataset's available items. When validated occupational-stress instruments are unavailable, the measure is labeled *perceived stress* rather than occupational stress, and work-related variables are treated as effect modifiers or covariates rather than substitutes for validated job-stress scales.

Primary outcomes are (i) **tinnitus status** (present/absent; bother modeled ordinally where available) and (ii) **audiometric hearing loss**, defined primarily as the better-ear speech-frequency pure-tone average (PTA at 0.5, 1, 2, 4 kHz) exceeding 25 dB HL, with high-frequency PTA (3, 4, 6 kHz) and asymmetry (interaural difference $\geq$15 dB at two adjacent frequencies) as prespecified alternatives. Secondary outcomes include self-reported hearing difficulty, hearing-aid use, and mental-health or quality-of-life correlates.

4.5 Statistical Analysis

All public-data analyses use survey-aware methods (Taylor-linearization variance estimation honoring weights, strata, and PSUs). Descriptive analyses estimate weighted prevalence of tinnitus and hearing impairment with 95% confidence intervals. Association analyses use weighted logistic, ordinal, or linear regression according to outcome type, with progressive adjustment: Model 1 (age, sex), Model 2 (+ education, occupation/employment, noise exposure), Model 3 (+ smoking, alcohol, sleep, cardiometabolic disease), and Model 4 (+ depressed mood/mental health). Results are reported as associations (odds ratios or mean differences) with 95% CIs, explicitly not as causal effects. Multiplicity is addressed by prespecifying primary hypotheses (Section 1) and controlling the false-discovery rate across secondary contrasts.

Missing data. Missingness is described by variable and mechanism. Primary analyses use multiple imputation by chained equations under a missing-at-random assumption (design variables included in the imputation model), pooled by Rubin's rules; complete-case analysis is reported as a sensitivity check.

Sample size and precision. Because KNHANES/NHANES sample sizes are fixed, we frame power as precision: with several thousand audiometry participants per pooled window, weighted prevalence is estimable to within a few percentage points, and moderate associations (OR $\approx$ 1.3–1.5) are detectable at conventional error rates. Precision, not post-hoc power, is reported for each primary estimate.

4.6 Prediction-Model Development and Validation

Two prediction targets—tinnitus status and audiometric hearing loss—are developed as prespecified, interpretable models. We first fit transparent baselines (survey-weighted logistic regression with pre-specified predictors), then gradient-boosted trees as a flexible comparator; the simpler model is preferred unless the complex model yields a clinically meaningful, calibrated improvement. Development uses KNHANES with repeated stratified cross-validation for internal validation and optimism correction. External validation uses NHANES, reporting both the frozen KNHANES model and a recalibrated variant (intercept/slope update). Performance is reported as discrimination (AUROC, AUPRC), calibration (calibration slope/intercept and plots, Brier score), and clinical utility (decision-curve analysis). Predictor importance is reported with SHAP values to preserve interpretability. All targets, predictors, and metrics are frozen before external validation (Table 4).

Fairness and subgroup evaluation. Discrimination and calibration are reported within prespecified subgroups (sex, age band, employment status, noise-exposure status, and unilateral/asymmetric hearing) to detect performance disparities. Subgroup gaps in calibration and error rates are treated as first-class results rather than afterthoughts.

4.7 Sensitivity Analyses

Prespecified sensitivity analyses test alternative hearing-loss definitions (25 vs. 20 dB HL; high-frequency PTA), age restrictions, exclusion of conductive patterns when tympanometry is available, alternative stress operationalizations, and stratification by employment status, noise exposure, sex, and unilateral/asymmetric hearing status.

4.8 Ethics and Governance

Stages 1–2 use publicly available, deidentified survey data and do not constitute human-subjects research requiring new consent; KNHANES and NHANES were approved by their respective review boards, and their use follows each provider's data-use terms. The Stage-3 clinical extension will obtain IRB approval at Ajou University Hospital, with a data-management plan covering deidentification, access control, and secondary-use consent. No identifiable patient data are placed in the public repository; only derived, deidentified, or synthetic artifacts are shared.

5 Open AI System and Research Acceleration

5.1 Motivation

A major barrier to audiology AI is the scarcity of harmonized, reusable clinical features. StressEar-AI addresses this by publishing a transparent feature schema, reproducible public-data code, and a clinician-verification protocol. This supports a broadly shared-benefit research model: investigators can reuse the pipeline, add local cohorts, and compare models without exposing identifiable patient data.

5.2 Model Roles and Prohibitions

The AI system has four restricted roles: (1) it harmonizes variable names and labels across public datasets; (2) it trains transparent baseline models for association and calibrated risk stratification; (3) in the clinical extension, an open medical language model (e.g., MedGemma) extracts structured fields from deidentified notes using a fixed JSON schema; and (4) it drafts explanations that cite only approved study documents or clinical guidelines. The system is explicitly prohibited from diagnosing an individual patient, assigning a definitive etiology, recommending or dosing steroids, or delaying urgent referral. These prohibitions are enforced in code and documented in a model card.

5.3 Deterministic Red-Flag Safety Layer

Before any probabilistic model output is surfaced, a deterministic rule layer screens for red flags derived from the SSNHL and tinnitus guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014): sudden or rapidly progressive hearing loss (especially unilateral), aural fullness with a discrete onset, single-sided tinnitus with asymmetry, vertigo, or focal neurologic signs. Any trigger overrides the model and returns an "urgent audiology/otolaryngology referral" message. The layer is evaluated primarily on recall (missed red flags are the costly error); its rules, versions, and test cases are unit-tested in the repository. This operationalizes the "golden-window" lesson from the motivating vignette: the framework must never let a probabilistic "low-risk, likely stress" estimate suppress timely referral.

5.4 Schema-Constrained Extraction and Governance

LLM extraction is constrained to a fixed schema with typed fields and controlled vocabularies; free-text generation is disallowed for structured outputs. Each extracted field is accompanied by a source span and a confidence flag, and every field is subject to clinician verification before entering analysis. Prompts, schema versions, model weights/version, temperature, and random seeds are logged. Extraction quality is reported as human-verified field accuracy, unsupported-output (hallucination) rate, and a clinician review of any potentially harmful output.

5.5 Robustness and Trustworthiness

Trustworthiness is pursued through eight design choices: (1) public-data baselines; (2) country-level external validation; (3) survey-weighted epidemiology; (4) interpretable models before complex models; (5) locked feature and outcome definitions; (6) clinician review of LLM-extracted variables; (7) subgroup calibration and error analysis; and (8) full run provenance (dataset versions, variable mappings, hyperparameters, prompts, seeds). Reproducibility is supported by pinned dependencies and deterministic seeds.

5.6 Open-Science Contribution

The repository includes executable analysis templates, a data dictionary, model cards, prompt templates, a red-flag test suite, and a completed reporting checklist. Because KNHANES files cannot be redistributed without official access procedures, the code specifies directory expectations and variable-mapping scripts rather than repackaging raw data. NHANES download helpers retrieve public files directly from CDC endpoints where available. A synthetic-data generator lets others exercise the full pipeline end-to-end without any restricted data, lowering the barrier to reuse and review.

6 Planned Results and Reproducible Scaffold

6.1 Planned primary results

No definitive epidemiological findings are reported in this pre-analysis manuscript. Per the prespecified aims (Section 1), the planned results will report: (i) weighted prevalence of tinnitus and hearing impairment in KNHANES with 95% CIs; (ii) adjusted associations between perceived stress and tinnitus across Models 1–4; (iii) adjusted associations between stress and audiometric thresholds, expected to attenuate after adjustment for age and noise (H1); (iv) subgroup analyses among employed middle-aged participants; (v) NHANES external-validation discrimination and calibration for the frozen and recalibrated models (H2); and (vi) subgroup fairness metrics. All tables will explicitly distinguish observed public-data results from prospective clinical-extension results.

6.2 Reproducible scaffold demonstration

To show that the pipeline runs end-to-end and produces correctly structured, TRIPOD+AI-aligned outputs, the repository ships a synthetic-data demonstration. These numbers are *illustrative artifacts of randomly generated data* and carry no epidemiological meaning; they exist only to verify wiring, metric computation, and reporting format. On the synthetic sample, the tinnitus-status target yields AUROC ≈ 0.64, AUPRC ≈ 0.49, and Brier ≈ 0.20, and the hearing-loss target produces an analogous metrics record (Table 4). When investigators substitute real KNHANES/NHANES files following the documented directory layout, the identical code path produces the design-consistent estimates specified above.

6.3 Planned AI-component reporting

Any LLM-extracted clinical field will be reported with human-verified accuracy, unsupported-output (hallucination) rate, and a clinician review of potentially harmful outputs. The deterministic red-flag layer will be reported by recall on a curated test set of red-flag and non-red-flag vignettes, with any missed red flag treated as a critical failure. Model cards will accompany each released model, documenting intended use, training data provenance, metrics by subgroup, and prohibitions.

7 Discussion

This AJA-targeted framework makes three deliberate commitments. First, it selects KNHANES as the primary public dataset because it is Korean, population-based, and relevant to the intended clinical setting, with NHANES for cross-national external validation. Second, it reframes AI as a reproducible research accelerator rather than a diagnostic authority, bounded by a deterministic safety layer. Third, it operationalizes reproducibility and shared benefit through fully open code, a data dictionary, model cards, and a synthetic-data path that lets anyone run the pipeline without restricted data.

7.1 Interpretation

The expected scientific contribution is *not* a claim that workplace stress directly damages hearing. Rather, StressEar-AI quantifies whether stress-related measures are consistently associated with tinnitus, hearing difficulty, and audiometric outcomes after accounting for noise exposure and health covariates. We anticipate a robust stress–tinnitus association and a weaker, noise-confounded stress–threshold association (H1). This asymmetry, if confirmed, has a clear clinical reading: stress may amplify tinnitus perception and burden more than it shifts cochlear thresholds, which aligns with tinnitus being partly a central, attention-modulated phenomenon.

7.2 Clinical contribution and the treatment window

The clinical contribution is an audiology-centered pathway that separates screening, referral, treatment, and rehabilitation. The framework's most consequential clinical stance is procedural rather than statistical: because SSNHL is a time-critical emergency (Chandrasekhar et al., 2019), the system is engineered so that a low predicted risk can never suppress urgent referral. The motivating vignette—resolved unilateral stress-associated tinnitus followed a year later by a delayed-care sudden loss in the other ear—illustrates precisely the failure mode the red-flag layer exists to prevent. Public data can identify population patterns and high-risk profiles; the Ajou University Hospital extension can then test treatment timing, SSNHL recovery, speech-in-noise outcomes, hearing-aid rehabilitation, and tinnitus burden longitudinally.

7.3 Comparison with prior work

Unlike single-country prevalence studies (Park et al., 2014; Joo et al., 2015), StressEar-AI prespecifies cross-national transportability testing and reports calibration and decision-curve utility, not discrimination alone. Unlike opaque clinical-AI demonstrations, it locks feature and outcome definitions before external validation, publishes model cards, and evaluates subgroup fairness as a primary result. This positions the framework as a reusable methodological baseline rather than a one-off result.

7.4 Limitations

Several limitations remain and temper interpretation. Stress measurement differs across datasets and may not capture validated occupational-stress constructs; we therefore label the exposure "perceived stress" when appropriate. Cross-sectional audiometry cannot establish temporal order, so no causal claim is made. Audiometry availability and item wording vary across survey cycles, constraining pooling. Tinnitus definitions and severity scales differ across sources. Public datasets lack detailed treatment timing, which only the prospective cohort can supply. Complex-survey design must be honored to avoid biased inference. Finally, open medical language models require local validation, privacy review, and clinical governance before any deployment; their outputs are confined here to research feature extraction and guideline-linked explanation, never to autonomous clinical decisions.

7.5 Future directions

Future work includes: adding longitudinal occupational cohorts to strengthen temporality; a neurophysiological substudy of tinnitus subtypes using open OpenNeuro/EEG datasets; audiogram-digitization to unlock legacy paper audiograms for retrospective analysis; and community extension of the harmonization schema to additional national health surveys, advancing a globally shared, reproducible evidence base.

8 Conclusion

StressEar-AI provides a public-data-first, clinically grounded, and openly reproducible framework for studying tinnitus, hearing loss, and stress-related exposures in employed adults.

KNHANES is the preferred development dataset, NHANES the preferred public external-validation dataset, and open medical AI models accelerate feature extraction and explanation only under schema constraints, a deterministic red-flag safety layer, and strict human verification. By prespecifying aims, honoring complex-survey design, reporting calibration and fairness, and releasing code, model cards, and a synthetic-data path, the framework improves reproducibility, robustness, and global usefulness while avoiding unsupported causal claims. Public data can establish associations and benchmark methods; the prospective clinical extension is required to study treatment timing and recovery—above all, to protect the narrow window in which sudden hearing loss is treatable.

Declarations

Author Contributions (CRediT): *Geunsik Lim*—Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing (original draft), Visualization. *Hyun Jo*—Clinical methodology (audiology/otolaryngology), Validation, Supervision of the planned clinical extension, Writing (review & editing). Both authors read and approved this draft.

Corresponding Author: Geunsik Lim, Sungkyunkwan University, Republic of Korea. Email: `leemgs@g.skku.edu`.

Funding: No external funding was received for this pre-analysis framework. Any funding for the prospective clinical extension will be disclosed at that stage.

Ethics Approval: Stages 1–2 use publicly available, fully deidentified survey microdata (KNHANES, NHANES) under each provider's data-use terms and do not constitute new human-subjects research. The Stage-3 clinical extension will obtain institutional review board approval at Ajou University Hospital before any data collection.

Reporting Guidelines: The observational analyses follow STROBE (von Elm et al., 2007); the prediction models follow TRIPOD+AI (Collins et al., 2024); the clinical extension will follow SPIRIT. A completed checklist is provided in the repository.

Data Availability: KNHANES data are available from the Korea Disease Control and Prevention Agency subject to official access procedures; NHANES public-use files are available from the U.S. CDC. The repository provides code, configuration, variable-mapping templates, and a synthetic-data generator, but does not redistribute restricted raw data.

Code Availability: All analysis code, configuration, model cards, prompt templates, and the reporting checklist are openly available at https://github.com/leemgs/hear-work-ai under an open-source license.

AI-Use Disclosure: Generative AI tools assisted in drafting and structuring this manuscript and the code scaffold; the authors reviewed and are responsible for all content. In the proposed study, open medical language models will be used only for research feature extraction and guideline-linked explanation from deidentified text, always with clinician verification before analysis and never for autonomous clinical decisions.

Conflicts of Interest: The authors declare no competing interests.

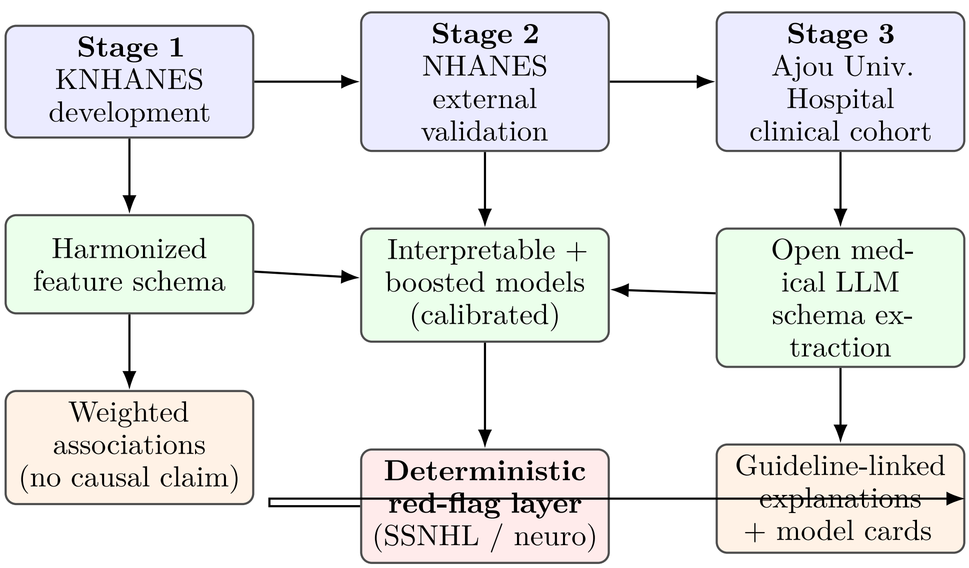

Figure 1: StressEar-AI conceptual framework. Public survey data (blue) feed a harmonized schema and calibrated, interpretable models (green); an open medical language model performs schema-constrained extraction in the clinical extension. A deterministic red-flag safety layer (red) can override probabilistic output to force urgent referral, so a low predicted risk never suppresses time-critical SSNHL care. Outputs (orange) are associations and guideline-linked explanations, not diagnoses.

Table 1: Selected public datasets for StressEar-AI

Dataset	Role	Strengths	Main limitations
KNHANES	Primary development dataset	Korean population relevance; audiometry in selected cycles; tinnitus and health variables; survey weights	Cross-sectional; stress variables may be perceived stress rather than validated occupational stress; raw data access procedures required
NHANES	External validation dataset	Public CDC files; audiometry and noise exposure in selected cycles; rich health covariates; geographic validation	U.S. population; cycle-specific hearing and tinnitus variables; complex survey design
OHHR / Zenodo audiology data	Secondary method testing	Audiology-specific measurements; useful for model benchmarking	May lack stress, occupation, or tinnitus variables
OpenNeuro tinnitus datasets	Optional neurophysiology substudy	Public neuroimaging or fNIRS data for tinnitus biomarkers	Not designed for occupational stress or longitudinal treatment outcomes

Table 2: Target harmonized variables and their availability in the primary public datasets. Availability is cycle-dependent; each derived variable records its source items and transformation in the version-controlled data dictionary.

Harmonized variable	Role	KNHANES	NHANES
Pure-tone thresholds (per ear, per frequency)	Outcome	Selected cycles	Selected cycles
Tinnitus status / bother	Outcome	Selected cycles	Selected cycles
Self-reported hearing difficulty	Outcome (2^{nd})	Yes	Yes
Hearing-aid use	Outcome (2^{nd})	Selected cycles	Selected cycles
Perceived stress / depressed mood	Exposure	Yes	Yes
Occupation / employment status	Modifier	Yes	Yes
Occupational + recreational noise	Covariate	Selected cycles	Yes
Sleep duration	Covariate	Selected cycles	Selected cycles
Smoking, alcohol	Covariate	Yes	Yes
Cardiometabolic disease	Covariate	Yes	Yes
Age, sex, education	Covariate	Yes	Yes
Survey weight, stratum, PSU	Design	Yes	Yes

Table 3: AI components and safety constraints

Component	Intended use	Safety constraint
Survey harmonization	Map KNHANES/NHANES variables to common schema	Human-readable mapping file and version control
Statistical baseline	Estimate adjusted associations and transparent risk scores	Survey-aware methods; no causal claims from cross-sectional data
MedGemma or similar open medical LLM	Extract fixed fields from deidentified clinical notes in future cohorts	Clinician verification; no diagnosis or treatment recommendation
Retrieval-augmented explanation	Draft evidence-linked explanations for researchers and clinicians	Curated sources only; unsupported outputs flagged
Red-flag rule layer	Identify sudden hearing-loss or neurologic warning signs	Deterministic referral rules override probabilistic model

Table 4: Prespecified evaluation metrics (frozen before external validation) and illustrative synthetic-data scaffold values. Synthetic values verify pipeline wiring only and carry no epidemiological meaning; real-data estimates replace them when KNHANES/NHANES files are supplied.

Dimension	Metric(s)	Synthetic scaffold (illustrative)
Discrimination	AUROC, AUPRC	Tinnitus: 0.64 / 0.49
Calibration	Calibration slope & intercept, Brier score, calibration plot	Tinnitus Brier: 0.20
Clinical utility	Decision-curve net benefit	Reported per target
Interpretability	SHAP predictor importance	Reported per target
Fairness	Subgroup AUROC & calibration (sex, age, employment, noise, asymmetry)	Reported per subgroup
LLM extraction	Human-verified accuracy, unsupported-output rate, harmful-output review	Clinical extension
Safety layer	Red-flag recall on curated vignette set	Target $\rightarrow$ 1.0

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